

Chapter 1: Ten Principles of Economics

Learning Objectives

- 1.1 Identify how individuals face trade-offs and how to analyze them.
- 1.2 Apply the meaning of opportunity cost and how to calculate it.
- 1.3 Identify marginal reasoning and calculate decisions at the margin.
- 1.4 Recognize how incentives affect people's behavior.
- 1.5 Identify why trade among people or nations can be good for everyone.
- 1.6 Recall why markets are a good, but not perfect, way to allocate resources.
- 1.7 identify factors that influence standards of living.
- 1.8 recall why there is an inverse relationship between inflation and unemployment.

Key Points

- The fundamental lessons about individual decision making are that people face trade-offs among alternative goals, that the cost of any action is measured in terms of forgone opportunities, that rational people make decisions by comparing marginal costs and marginal benefits, and that people change their behavior in response to the incentives they face.
- The fundamental lessons about economic interactions among people are that trade and interdependence can be mutually beneficial, that markets are usually a good way of coordinating economic activity, and that the government can potentially improve market outcomes by remedying a market failure or by promoting greater economic equality.
- The fundamental lessons about the economy as a whole are that productivity is the ultimate source of improving living standards, that growth in the quantity of money is the ultimate source of inflation, and that society faces a short-run trade-off between inflation and unemployment.

What Economics is All About

- What is **economics**?
 - Money, markets, stocks and bonds?
- **Definition:** the study of how society manages its scarce resources.
- Economists study how all economic units in society react to the problem of scarcity

Three items at the heart of economic analysis:

- **Scarcity** – the limited nature of society's resources
- **Choice** – alternative options for people to pursue
- **Marginal** – additional or incremental

How People Make Decisions: Principle 1

Principle #1 : People Face Tradeoffs

- Going to class vs. going to work or sleeping
- Going to a party or a night out vs. studying for a midterm
- Deciding between two job openings

- Society faces a tradeoff as well

Efficiency vs. Equality

- **Efficiency** – when society gets the most from its scarce resources
- **Equality** – when prosperity is distributed uniformly among society's members
- Which is better?

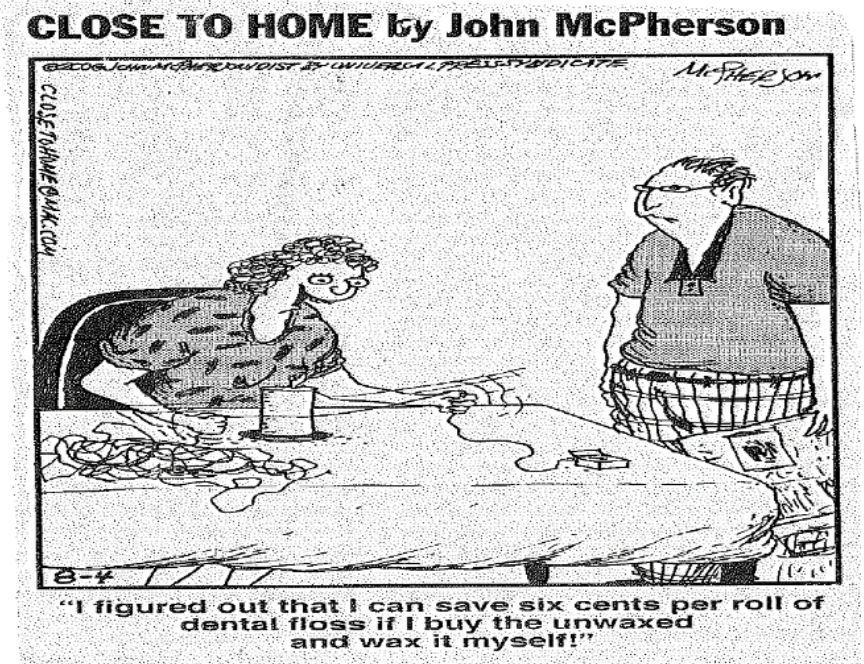
Share of Income across Countries



How People Make Decisions: Principle 2

Principle #2: The Cost of Something is What You Give Up to Get It

- “There ain’t no such thing as a free lunch”
- **Opportunity Cost** – highest valued forgone alternative
- What’s the opportunity cost of attending class?
- What’s the opportunity cost of attending college?
- Why go to college if it costs you this much to do so?
- Because median wages for college grads is \$49,000



Activity – Opportunity Costs

- What's the opportunity cost of going to college?

Item	Cost
Tuition	\$13,000
Books	\$1,000
Room/Board	\$10,000
Forgone Wages	\$45,000
Total (1 year)	\$24,000
Total (4 years)	\$96,000

Activity – Opportunity Costs

- The direct cost of going to college is \$24,000 for one year and \$96,000 for four years.
- The indirect cost (what you gave up to go to college) is \$45,000 for one year and \$180,000 for four years.
- The opportunity cost of going to college is \$69,000 for one year and \$276,000 for four years.

How People Make Decisions: Principle 3

Principle #3: Rational People Think at the Margin

- **Rational** – Systematic and purposeful
- Recall: **Marginal** – additional or incremental
- A rational person compares the marginal benefits (MB) to marginal cost (MC)
 1. If **MB** > **MC** the rational person pursues the activity
 2. If **MC** > **MB** the rational person does not pursue the activity
- Ex: You sit in class for 1 hour and your utility (satisfaction from consumption) has decreased. You decide that another minute of lecture causes the marginal cost to your satisfaction to outweigh the marginal benefit of learning. What do you do?

Activity – Should You Fix the Car?

You are selling your 2015 Nissan Rogue. You have already spent \$1,000 on repairs.

At the last minute, the transmission dies. You can pay \$1,500 to have it repaired or sell the car “as is.”

In each of the following scenarios, should you have the transmission repaired?

1. Blue book value (what you could get for the car) is \$10,500 if transmission works, \$7,000 if it doesn't.
2. Blue book value is \$10,500 if transmission works, \$9,500 if it doesn't.

Activity – Should You Fix the Car?

Scenario 1

- Marginal benefit = $\$10,500 - \$7,000 = \$3,500$
- Marginal cost = $\$1,500$
- It's worthwhile to have the transmission fixed: $MB > MC$

Scenario 2

- Marginal benefit = $\$10,500 - \$9,500 = \$1,000$
- Marginal cost = $\$1,500$
- It's not worthwhile to fix the transmission: $MC > MB$

How People Make Decisions: Principle 4

Principle #4: People Respond to Incentives

- **Incentive:** something that induces a person to act, i.e. the prospect of a reward or punishment.
- 1. When gas prices rise, consumers buy more hybrid cars and fewer gas guzzling SUVs.
 - *Unintended consequence* – more drivers on the road because of better fuel economy
- 2. When lectures bore you to death you respond by not going to class anymore
 - *Unintended consequence* – you have a greater risk of failing the course
- 3. When the government extends unemployment insurance to help the unemployed
 - *Unintended consequence ?*
- 4. When a panic ensues and all the n-95 masks are sold out at stores and are being sold by third parties for a 400% mark-up
 - *Unintended consequence?* Medical workers in hospitals don't have access to masks and are now at a greater risk of contracting the virus.

Activity: Incentives and Unintended Consequences

- Choose one of the following:
 - Government stimulus checks
 - Loyalty rewards programs
 - Surge pricing on Uber or DoorDash
 - Extended unemployment benefits
- What incentive was created?
- What was the intended result of the incentive?
- What unintended consequences emerged?
- Was the incentive effective? Would you redesign it?

Activity: Incentives and Unintended Consequences

Government Stimulus Checks

Incentive created: Put money directly into households' hands to encourage spending and stabilize demand during an economic downturn.

Intended result: Boost consumer spending, prevent recessionary collapse, and help families meet short-term expenses.

Unintended consequences:

- Some recipients saved or invested rather than spent, limiting the short-run demand effect.
- Temporary labor shortages as some workers delayed returning to low-wage jobs.
- Inflationary pressures when aggregate demand rebounded faster than supply.

Effectiveness: Moderately effective—stimulus kept consumption and GDP from falling further, but timing and scale fueled later inflation.

Redesign idea: Phase payments gradually or tie them to income loss rather than universal checks.

Incentives and Organ Transplants

- Where do we draw the line between self-interest and morality?
- Incentives are invaluable as they help determine our actions.
- Are some things too taboo to really use incentives and the market?
 - Organ transplants, legalizing prostitution, same sex marriages, carbon taxes, etc.
- Let's consider organ transplants.
- Most economists agree that paying people for kidneys would increase the supply of kidneys available and limit the amount of deaths that occur among those waiting.
- The issue is that some may find this morally reprehensible, as they argue poorer people respond to the monetary incentive because they are the most in need, but may regret the decision later in life.
- Alvin Roth, the 2012 Nobel Laureate in Economics, whose idea has been put into practice, finds a way around this moral dilemma. In essence, the idea creates an exchange market between two pairs of people who are willing to donate a kidney but are incompatible with the other member in their respective pairing.
 - In essence, people are still responding to incentives.

How People Interact: Principle 5

Principle #5: Trade Can Make Everyone Better Off

- Rather than being self-sufficient, people can specialize in producing one good or service and exchange it for other goods.
- Countries also benefit from trade and specialization:
 - Get a better price abroad for goods they produce
 - Buy other goods more cheaply from abroad than could be produced at home
- **Caveat: Few instances arise when trade doesn't make you better off**

How People Interact: Principle 6

Principle #6: Markets are Usually a Good Way to Organize Economic Activity

- **Market** – a group of buyers and sellers
 - What goods to produce?
 - How to produce them?
 - How much of each to produce?
 - Who gets them?
- **Market Economy** – allocates resources through the decentralized decisions of many households and firms as they interact in markets.
 - Is socialism better than capitalism? Read Marx/Engels, Smith, von Mises
- Adam Smith and *The Wealth of Nations* (1776)
- **Invisible hand**
 - *Every individual... neither intends to promote the public interest, nor knows how much he is promoting it... he intends only his own security; and by directing that industry in such a manner as its produce may be of the greatest value, he intends only his own gain, and he is in this, as in many other cases, led by an invisible hand to promote an end which was no part of his intention.*

How People Interact: Principle 7

Principle #7: Governments Can Sometimes Improve Market Outcomes

- Enforcement of **property rights** – legal claim to private property
- **Market Failure** – when the market fails to allocate society's resources efficiently

Occurs in three ways:

1. **Externalities** – when the production or consumption of a good affects bystanders
2. **Public Good** – a good that once produced is free for everyone to consume
3. **Market Power** – a single buyer/seller with substantial influence on market price

Corrective measures:

1. Taxes/subsidies
2. Provide them
3. Regulation

What is the Optimal Amount?

- Back to efficiency and equity...
- “Any Pareto efficient allocation of resources can be achieved as a competitive equilibrium provided that initial endowments of resources are appropriately redistributed (through lump-sum transfers), and assuming no market imperfections.”
- **Efficiency** is handled by markets.
- **Equity** (fairness) can be addressed through government intervention (via redistribution), without distorting the efficiency of the market.

Luxury Island and Redistribution

Imagine a society with two islands:

- **Island A** has all the capital, infrastructure, and technology. Its residents are highly productive.
- **Island B** has only basic resources and manual labor. Its residents work hard but earn very little due to their limited means.

Trade between the two islands is open and efficient. Over time, Island A becomes vastly wealthier. Island B's people can buy food and goods but have little hope of ever reaching Island A's quality of life—even though the markets are functioning efficiently.

- Now, suppose a global governing body offers a one-time wealth redistribution—a lump-sum transfer of resources from Island A to Island B. After this, the free market is left to operate without interference.

Luxury Island and Redistribution

1. If markets are efficient but produce extreme inequality, should governments intervene to redistribute endowments?
2. Does a fair starting point justify letting markets run freely afterward? Or should we always balance equity and efficiency simultaneously?
3. What forms of redistribution (education, healthcare, direct cash transfers) are least distortionary, and do they actually preserve market efficiency?

How the Economy as a Whole Works: Principle 8

Principle #8: A Country's Standard of Living Depends on Its Ability to Produce Goods and Services

- The most important determinant of living standards:
 - **Productivity** – the amount of goods and services produced per unit of labor
 - Capital, human capital, technology and institutions drive productivity
 - The more productive someone is the higher their real income
 - Other reasons explain higher levels of real income
 - Natural resources, geographic advantages, etc.
- Higher levels of production mean higher levels of income
- Other factors (e.g., labor unions, competition from abroad) have far less impact on living standards.

Mini Case: South Korea's Rise in Productivity

In the 1950s, South Korea had an economy primarily based on agriculture and war recovery efforts. Fast forward to today, and it's a global leader in electronics, automobiles, and shipbuilding.

Why the transformation?

- **Massive investment in education:** Literacy, STEM training, and a strong work ethic.
 - **Technology and innovation:** Companies like Samsung, Hyundai, and LG now dominate global markets.
 - **Infrastructure:** Efficient transportation, internet access, and industrial zones.
 - **Trade openness:** Export-driven strategies connected Korea to global markets.
- Result:** Higher productivity → higher wages → drastically improved standard of living for the average citizen.
- South Korea is a great example of how productivity growth translates directly into real income growth.

How the Economy as a Whole Works: Principle 9

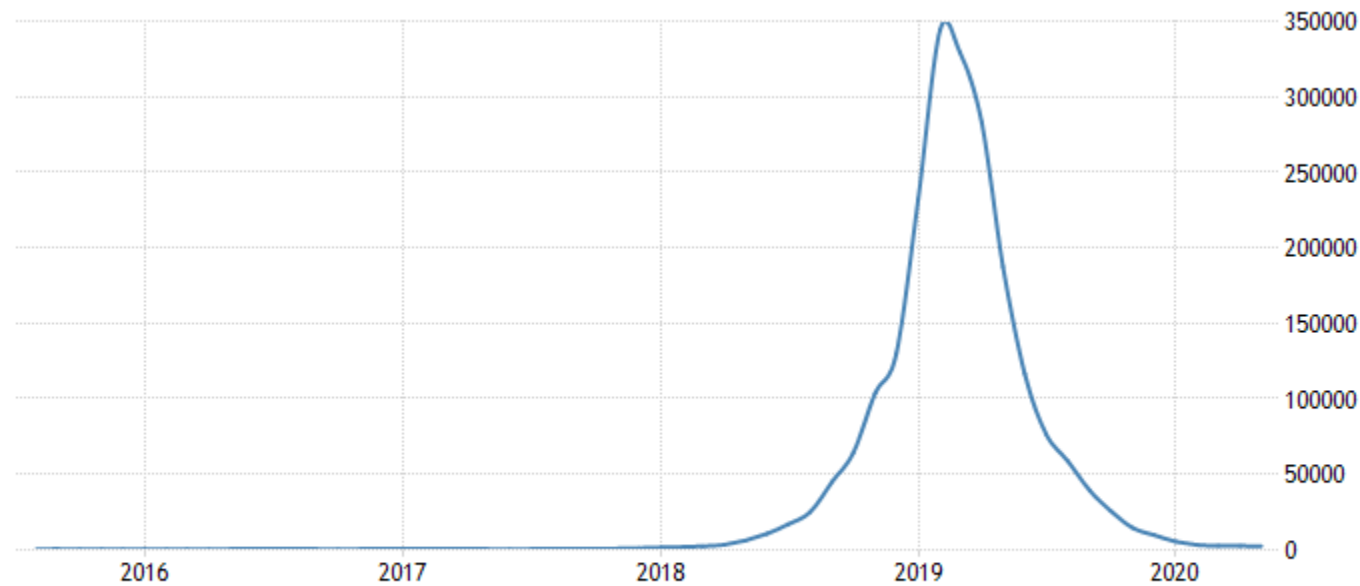
Principle #9: Prices Rise When the Government Prints Too Much Money

Inflation: increases in the general level of prices.

- In the long run, inflation is almost always caused by excessive growth in the quantity of money, which causes the value of money to fall.
- The faster the government creates money, the greater the inflation rate.
- The Federal Reserve just engaged in \$700B in quantitative easing and near 0% interest rates

Hyperinflation in Venezuela

- Venezuela's problem:
 - “I was recently in the city of Cucuta on Colombia's border with Venezuela, and I met a man creating paper sculptures for sale by the side of the road. Jorge Cordero is 19 years old. He weaves and braids paper into swans, purses, owls, trucks. And all of the paper is Venezuelan currency - bolivars.”
 - [From How Venezuela's Currency Became so Worthless](#)



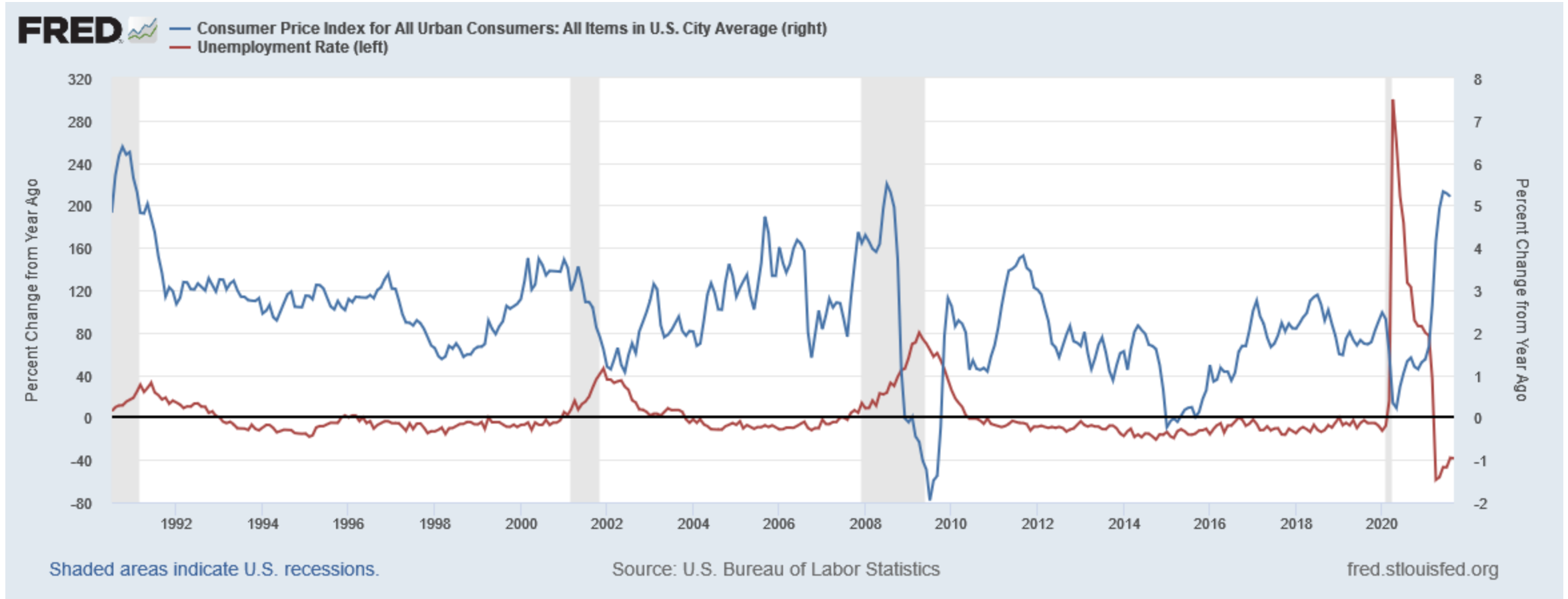
SOURCE: TRADINGECONOMICS.COM | BANCO CENTRAL DE VENEZUELA

How the Economy as a Whole Works: Principle 10

Principle #10: Society Faces a Short-run Tradeoff Between Inflation and Unemployment

- In the short-run (1–2 years), many economic policies push inflation and unemployment in opposite directions.
- Other factors can make this tradeoff more or less favorable, but the tradeoff is always present.
- Currently we are around 5.3% inflation Y/Y and 4.8% unemployment
 - Other inflation categories: food 3.7%, food at home 3%, gasoline 33.2%, used cars and trucks 31.9%
 - Other unemployment categories: labor force 161.4 million, participation rate 61.7%, not in labor force 100.4 million

CPI and Unemployment Rate: 1990 - 2021



Key Takeaways

How People Make Decisions (Principles 1–4)

- **Scarcity and choice** are at the core of economics; every decision involves trade-offs.
- The **opportunity cost** of an action is the value of the next best alternative you give up.
- **Rational thinkers** compare *marginal benefit (MB)* to *marginal cost (MC)*—act only when $MB \geq MC$.
- **Incentives** matter—people respond to rewards and penalties, often with **unintended consequences**.

How People Interact (Principles 5–7)

- **Trade** and specialization make all parties better off.
- **Markets** usually coordinate decisions efficiently through prices—the “invisible hand.”
- **Government** can improve outcomes when markets fail or equity goals require redistribution.

How the Economy as a Whole Works (Principles 8–10)

- **Productivity** is the main driver of living standards.
- **Excess money growth** causes **inflation**; stable prices depend on monetary discipline.
- In the **short run**, policy faces a **trade-off between inflation and unemployment** (Phillips curve).

Economics provides a framework for understanding choices—from everyday personal decisions to national policy—by focusing on incentives, trade-offs, and how resources move through the system.

Knowledge Check

1. You decide to take a weekend trip that costs \$400. The job you skip that weekend would have paid \$300. What is your total opportunity cost of the trip?
2. You plan to study two more hours for your exam. The first hour will likely raise your grade by 5 points; the second, by 1 point. What concept helps you decide whether to study the second hour?
3. When cities impose rent controls to make housing affordable, what unintended consequence often occurs?
4. The “invisible hand” described by Adam Smith refers to:
5. Which statement best summarizes the long-run relationship between money and prices?

Knowledge Check Solutions

1. \$700
2. Marginal analysis
3. The quality and quantity of housing decrease
4. Self-interest leading to efficient market outcomes
5. Money growth leads to inflation overtime